

PROPERTY AGGREGATION & EXPOSURE INTELLIGENCE PLATFORM

Geospatial Analytics | Exposure Management | Underwriting Decision Support

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EXECUTIVE SUMMARY

Business Challenge

Homeowners insurance underwriters must evaluate not only individual property risk but also portfolio concentration and geographic accumulation.

Existing reviews relied heavily on spreadsheets, individual address lookups, and manual portfolio analysis, making it difficult to identify exposure concentrations and aggregation hotspots.

Project Objective

Develop an interactive geospatial platform that enables underwriters to visualize insured locations, identify concentration zones, monitor portfolio accumulation, and support underwriting decisions through location-based exposure intelligence.

Key Benefits

- Improved visibility into geographic concentration
- Faster portfolio review
- Enhanced catastrophe exposure awareness
- Better underwriting decision support
- Centralized exposure monitoring

PROCESS TRANSFORMATION

Before

Policy Data



Spreadsheet Review



Address Lookup



Manual Aggregation Analysis



Underwriting Decision

Challenges

- Limited geographic visibility
- Time-consuming analysis
- Difficult concentration monitoring
- Fragmented exposure information

After

Policy Portfolio



Exposure Aggregation Engine



Geospatial Analysis



Interactive Mapping Platform

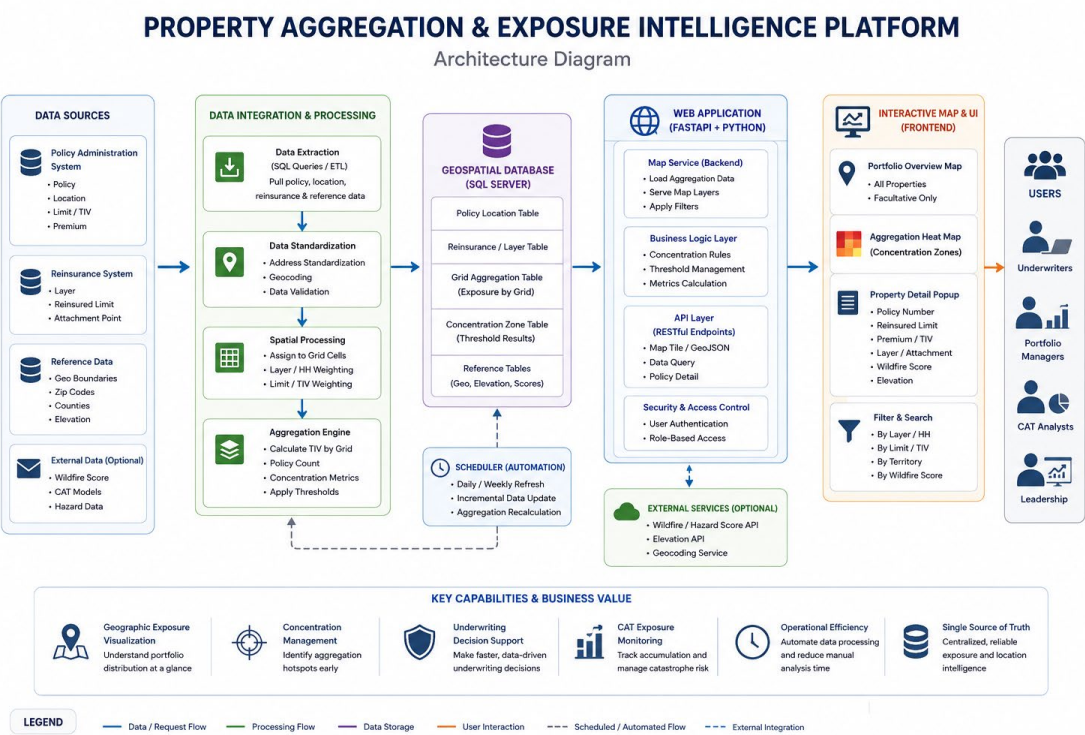


Underwriting Decision Support

Benefits

- Visual concentration monitoring
- Faster risk assessment
- Centralized portfolio view
- Improved accumulation management

SOLUTION ARCHITECTURE



Suggested Architecture

Policy Database



SQL Data Extraction



Exposure Aggregation Engine



Grid-Based Concentration Analysis



Geospatial Mapping Layer



Interactive Dashboard



Underwriting Team

Core Components

- Policy Data Integration
- Exposure Aggregation Logic

- Geographic Grid Analysis
- Concentration Scoring
- Interactive Mapping
- Portfolio Monitoring

GEOSPATIAL EXPOSURE DASHBOARD

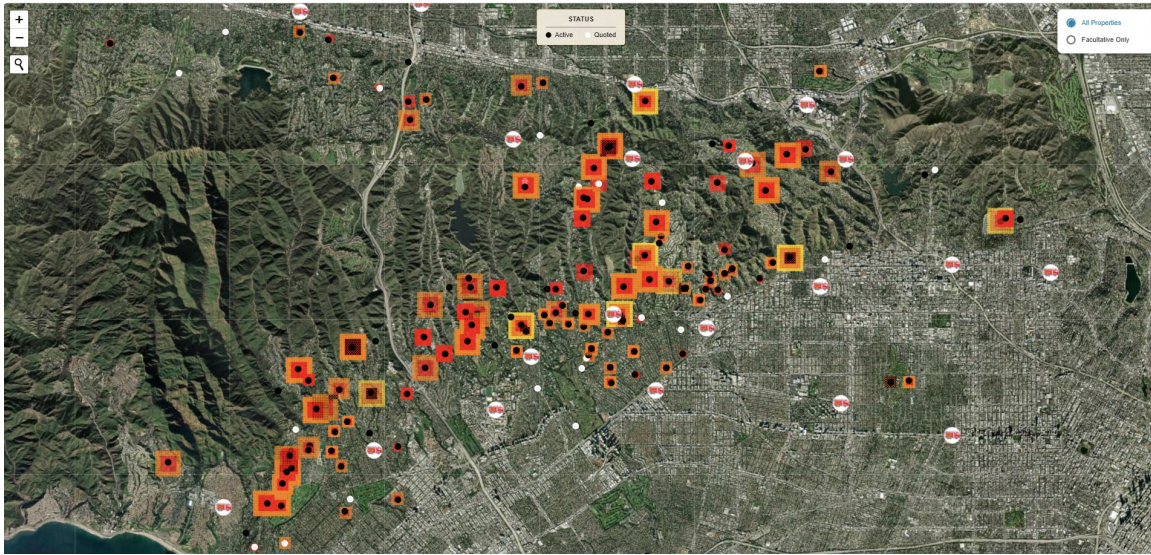


Figure 1. Portfolio Exposure Overview

The platform displays:

- Active Properties
- Quoted Properties
- Facultative Placements
- Geographic Concentration Zones
- Portfolio Exposure Distribution



Figure 2. Exposure Concentration Analysis

The system highlights:

- High-concentration areas
- Aggregation hotspots
- Property-level details
- Exposure metrics
- Underwriting intelligence

TECHNICAL IMPLEMENTATION & BUSINESS IMPACT

Technologies

Backend

- Python
- Pandas
- SQL Server

Geospatial Analytics

- Folium
- Geographic Grid Analysis
- Spatial Aggregation Logic

Data Sources

- Policy Portfolio Data
- Exposure Values
- Underwriting Information

Business Impact

- Improved visibility into geographic risk concentration
- Supported underwriting accumulation management
- Reduced manual portfolio review effort
- Enhanced catastrophe exposure monitoring
- Enabled faster underwriting decisions
- Consolidated property exposure data into a single view